

# URL Shortener — API Endpoints

**Project:** URL Shortener  
**Technology:** Python, FastAPI, SQLAlchemy, SQLite, Pydantic  
**Location:** /home/dominic/Documents/fastapiurl  
**Report #:** 3/20

## Endpoints Overview

| Method | Path                  | Description              |
|--------|-----------------------|--------------------------|
| POST   | /shorten              | Create short URL         |
| GET    | / {short_code}        | Redirect to original URL |
| GET    | / {short_code} /stats | Get click stats          |
| GET    | / {short_code} /json  | Get JSON info            |
| GET    | / {short_code} /html  | Get HTML page            |
| GET    | /list                 | List all URLs            |
| DELETE | / {short_code}        | Delete URL               |

## Complete API Reference

### Endpoint Table

| Method                    | Path | Auth | Request | Response | Description |
|---------------------------|------|------|---------|----------|-------------|
| POST /auth/register       | TODO | TODO | TODO    |          |             |
| POST /auth/login          | TODO | TODO | TODO    |          |             |
| GET /auth/me              | TODO | TODO | TODO    |          |             |
| POST /urls/shorten        | TODO | TODO | TODO    |          |             |
| GET / {short_code}        | TODO | TODO | TODO    |          |             |
| GET /urls/stats/{code}    | TODO | TODO | TODO    |          |             |
| GET /urls/list            | TODO | TODO | TODO    |          |             |
| DELETE /urls/{code}       | TODO | TODO | TODO    |          |             |
| PATCH /urls/{code}/toggle | TODO | TODO | TODO    |          |             |

# Request/Response Examples

## Example: Successful Request

```
// POST /auth/login
// Request:
{"email": "user@example.com", "password": "..."}
// Response (200):
{"access_token": "eyJ...", "token_type": "bearer"}
```

## Example: Error Response

```
// Response (401):
{"detail": "Invalid credentials"}
```

## Example: Paginated List

```
// GET /urls/list?skip=0&limit=10
// Response (200):
{"items": [...], "total": 42, "skip": 0, "limit": 10}
```

# Status Codes Used

| Code | Meaning        | Usage                              |
|------|----------------|------------------------------------|
| 200  | OK             | Successful GET, POST, PATCH        |
| 201  | Created        | Successful POST (resource created) |
| 204  | No Content     | Successful DELETE                  |
| 400  | Bad Request    | Validation error                   |
| 401  | Unauthorized   | Missing/invalid JWT                |
| 403  | Forbidden      | Not resource owner                 |
| 404  | Not Found      | Resource doesn't exist             |
| 409  | Conflict       | Duplicate resource                 |
| 422  | Unprocessable  | Pydantic validation failure        |
| 429  | Rate Limited   | Too many requests                  |
| 500  | Internal Error | Server-side failure                |